

Rick Farouni

[Personal Page](#) | <https://github.com/rfarouni> | tfaruni@hse.ru | [Google Scholar](#)

EXPERIENCE

Associate Professor <i>The Higher School of Economics, Faculty of Computer Science</i>	09/2026-Present Moscow, Russia
Bioinformatics Scientist <i>The Swiss Federal Institute of Technology in Zurich (ETH Zurich)</i>	10/2020-08/2026 Basel, Switzerland
Postdoctoral Researcher in Computational Biology <i>McGill University-Génomique Québec Innovation Centre</i>	06/2018-09/2020 Montreal, Canada
Postdoctoral Researcher in Computational Biology <i>Harvard Medical School-Massachusetts General Hospital</i>	06/2017-05/2018 Massachusetts, USA
Graduate Teaching Assistant in Statistics <i>The Ohio State University</i>	08/2013-05/2017 Ohio, USA

EDUCATION

PhD in Quantitative Psychology (latent variable modeling) <i>The Ohio State University</i>	Ohio, USA 08/2012-05/2017
Dissertation: <i>An Overview of Probabilistic Latent Variable Models with an Application to the Deep Unsupervised Learning of Chromatin States</i> . Farouni, T. (2017). https://etd.ohiolink.edu/acprod/odb_etd/ws/send_file/send?accession=osu1492189894812539&disposition=inline	
Master of Science in Mathematical Statistics <i>The Ohio State University</i>	Ohio, USA 08/2012-05/2015
Master's Degree in Psychometrics <i>The Ohio State University</i>	Ohio, USA 08/2012-12/2014
Thesis: <i>Latent Variable Models of Categorical Responses in the Bayesian and Frequentist Frameworks</i> . Farouni, T. (2014). Manuscript available at https://etd.ohiolink.edu/etd.send_file?accession=osu1412374136&disposition=inline	
Bachelor's Degree in Psychology <i>The Pennsylvania State University</i>	Pennsylvania, USA 01/2011-05/2012

Awards and Fellowships

Graduate Student Conference Presentation Award

The Ohio State University 2015

The Center for Cognitive and Brain Sciences Summer Graduate Fellowship

The Ohio State University 2015

Project Proposal: Across-Subject Predictive Modeling of fMRI BOLD Responses to Faces using a sparse Bayesian Group Factor Analysis Model (<https://rfarouni.github.io/assets/posters/jsm2015.pdf>)

The Social and Behavioral Sciences Summer Fellowship

The Ohio State University 2014

University Fellowship

The Ohio State University 2012

PUBLICATIONS

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<https://orcid.org/0000-0002-3454-2946>

H-index: 12; Total Citation Count: 4,000+

Journal Papers (first author/co-author)

- Lazure, F.†, **Farouni, R.**†, Sahinyan, K., Blackburn, D.M., Hernández-Corchado, A., Perron, G., Lu, T., Osakwe, A., Ragoussis, J., Crist, C. and Perkins, T.J., Jahani-Asl, A., Najafabadi, H.S., & Soleimani, V.D. (2023). Transcriptional reprogramming of skeletal muscle stem cells by the niche environment. *Nature Communications*.
<https://www.nature.com/articles/s41467-023-36265-x>.
Website: https://csglab.github.io/transcriptional_reprogramming_muscle_cells/.
- **Farouni, R.**, Djambazian, H., Ferri, L. E., Ragoussis, J., & Najafabadi, H. S. (2020). Model-based analysis of sample index hopping reveals its widespread artifacts in multiplexed single-cell RNA-sequencing. *Nature Communications*.
<https://www.nature.com/articles/s41467-020-16522-z>
Website: <https://csglab.github.io/PhantomPurgeR/>.
- Clement, K.†, **Farouni, R.**†, Bauer, D. E., & Pinello, L. (2018). Design and analysis of unique molecular identifiers for deep amplicon sequencing. *Bioinformatics*.
<https://doi.org/10.1093/bioinformatics/bty264>
- Pinello, L.†, **Farouni, R.**†, & Yuan, G-C. (2018). Haystack: systematic analysis of the variation of epigenetic states and cell-type specific regulatory elements. *Bioinformatics*.
<https://doi.org/10.1093/bioinformatics/bty031>

- Baskin, E.†, **Farouni, R.**†, & Mathè, E. (2017). ALTRE: workflow for defining ALTerred Regulatory Elements using chromatin accessibility data. *Bioinformatics*. <https://doi.org/10.1093/bioinformatics/btx386>
- Farouni, R. (2017). A Contemporary Overview of Probabilistic Latent Variable Models. *arXiv* preprint. Preprint available at <https://arxiv.org/abs/1706.08137>

Journal Papers (middle author)

- Belli, O., Karava, K., **Farouni, R.**, & Platt, R. J. (2024). Multimodal scanning of genetic variants with base and prime editing. *Nature Biotechnology*.
 - Santinha, A. J., Klingler, E., Kuhn, M., **Farouni, R.**, Lagler, S., Kalamakis, G., Lischetti, U., Jabaudon, D., & Platt, R. J. (2023). Transcriptional linkage analysis with in vivo AAV-Perturb-seq. *Nature*.
 - Schmidt, F., Zimmermann, J., Tanna, T., **Farouni, R.**, Conway, T., Macpherson, A.J., & Platt, R.J. (2022). Noninvasive assessment of gut function using transcriptional recording sentinel cells. *Science*.
 - Anderson-Trocmé, L., **Farouni, R.**, Bourgey, M., Kamatani, Y., Higasa, K., Seo, J., Kim, C., Matsuda, F., & Gravel, S. (2019). Legacy Data Confounds Genomics Studies. *Molecular Biology and Evolution*.
 - Seruggia, D., Oti, M., Tripathi, P., Canver, M. C., LeBlanc, L., Di Giammartino, D. C., Bullen, M.J., Nefzger C.M., Sun, Y.B.Y., **Farouni, R.**, Polo, J.M., Pinello, L., Apostolou, E., Kim J., Orkin, S.H., & Das, P.P. (2019). TAF5L and TAF6L Maintain Self-Renewal of Embryonic Stem Cells via the MYC Regulatory Network. *Molecular Cell*.
 - Clement K., Rees H., Canver, M.C., Gehrke J. M., **Farouni, R.**, Hsu, J.Y., Cole, M., Liu D.R., Joung J.K., Bauer, D.E., & Pinello, L. (2019). CRISPResso2 provides accurate and rapid genome editing sequence analysis. *Nature Biotechnology*.
 - Hsu, J. Y., Fulco, C.P., Cole, M., Canver, M.C., Pellin D., Sher, F, **Farouni, R.**, Clement K., Biasco L., Engreitz, J.M., Lander, E.S., Joung J.K., Bauer, D.E., & Pinello, L. (2018). CRISPR-SURF: Exploratory and interactive software for analyzing CRISPR-based tiling screens. *Nature Methods*.
 - Grüning, B., Dale, R., Sjödin, A., Rowe, J., Chapman, B. A., Tomkins-Tinch, C. H., The Bioconda Team, & Köster, J.(2018). Bioconda: A sustainable and comprehensive software distribution for the life sciences. *Nature Methods*. (Team Contributor)
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Software Development

- *generative_bioseq_design*: Generative deep learning framework for biological sequence design, targeting AAV9 capsid engineering. Combined PyTorch Lightning training infrastructure with attention-based classifiers and interpretability analysis. GitHub Repo: https://github.com/rfarouni/generative_bioseq_design
 - *PhantomPurgeR*: The optimal purging of phantom molecules by the robust estimation of the sample index hopping rate in multiplexed droplet-based single-cell RNA-seq data. GitHub Repo: <https://csglab.github.io/PhantomPurgeR/>.
 - *Haystack*: A Python bioinformatics pipeline for the identification of genomic regions of epigenetic variability across different cell-types, cell-type specific cis-regulatory elements, and their associated transcription factors. GitHub Repo: https://github.com/pinellolab/haystack_bio.
 - *ALTRE*: A Workflow for Identifying ALTerred Regulatory Elements using Chromatin Accessibility Data. GitHub Repo: <https://github.com/Mathelab/ALTRE>.
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LANGUAGES

English, Arabic, and Russian.